



Hands-On Activity: Generate a chart from a spreadsheet



Activity Overview

At this point in the course, you've explored how spreadsheets can be used to support data-driven business decisions. In this activity, you will use a spreadsheet to make a simple chart. By the time you're done, you'll understand how to create a simple graphical representation of information. Data analysts use this skill to make reports, presentations, infographics, and more engaging and accessible.

Scenario

Review the following scenario. Then complete the step-by-step instructions.

The scenario: Analyze patterns in monthly sales

To help determine optimal inventory and staffing levels, your company has asked you to analyze total sales trends for the last three years. Because your firm relies on tourists for the bulk of its sales, leaders know that demand for inventory and staffing requirements vary based on the tourist season. They have asked you to identify peak months to help with forecasting requirements for next year. To do this, you'll create a chart.

Step-By-Step Instructions

Working with spreadsheets

Follow the instructions to complete each step of the activity. Then answer the questions at the end of the activity before going to the next course item.

Step 1: Access the template



The monthly sales spreadsheet

To use the spreadsheet for this course item, select the link below, then select the “Use Template” button to open your own version of the spreadsheet.

Link to template: [Monthly sales](#)

OR

If you don’t have a Google account, download the template directly by selecting the attachment below.

[Monthly sales](#)

[XLSX File](#)

Save the spreadsheet using a preferred file naming convention and store it in a folder to help you stay organized.

Step 2: Open the spreadsheet

If you haven’t done so already, open the [Monthly sales](#) spreadsheet.

Step 3: Review the data

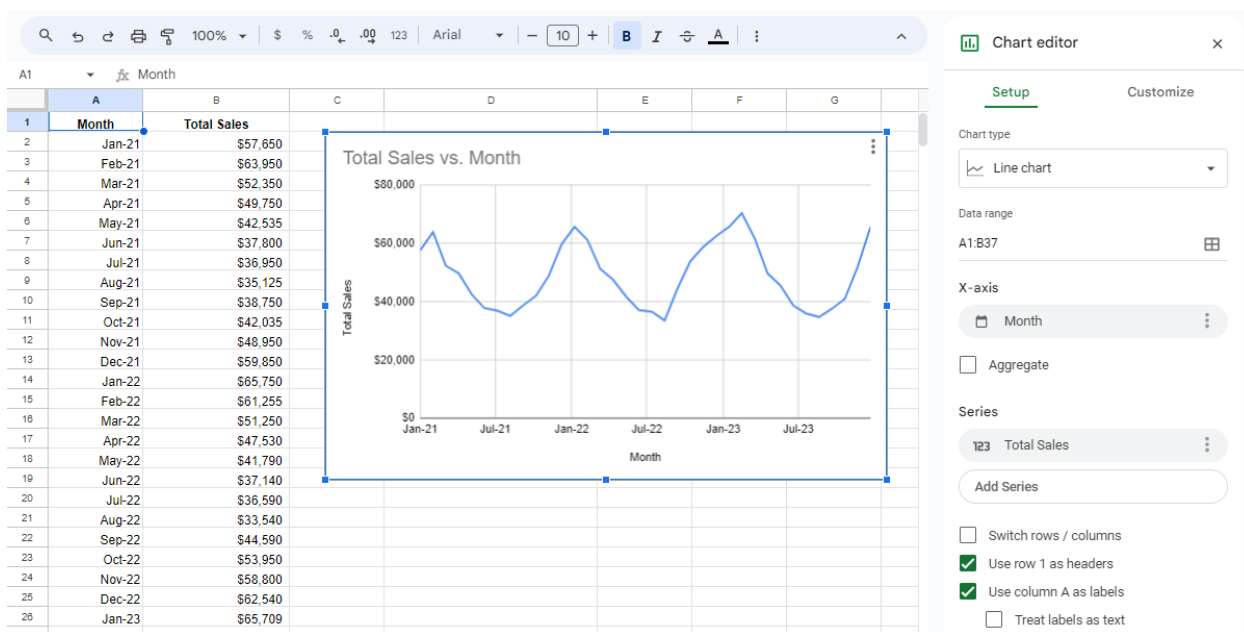
This spreadsheet contains total sales data for each month beginning in January 2021 and ending in December 2023. A portion of the spreadsheet is shown below.

A1	fx	Month
	A	B
1	Month	Total Sales
2	Jan-21	\$57,650
3	Feb-21	\$63,950
4	Mar-21	\$52,350
5	Apr-21	\$49,750
6	May-21	\$42,535
7	Jun-21	\$37,800
8	Jul-21	\$36,950
9	Aug-21	\$35,125
10	Sep-21	\$38,750
11	Oct-21	\$42,035
12	Nov-21	\$48,950
13	Dec-21	\$59,850
14	Jan-22	\$65,750

Step 4: Create a chart

To make it easier to understand monthly sales trends, use these steps to create a chart:

1. Select cell **A1**.
2. Choose **Insert > Chart**. Google Sheets inserts a chart to the right of the data. By default, Google Sheets creates a line chart.
3. Notice that Google Sheets generated a title automatically for the chart, “Total Sales vs. Month,” and added labels of “Total Sales,” for the vertical axis, and “Month” for the horizontal axis. It also automatically opens the **Chart editor** panel on the right side of the window, with the **Setup** tab opened.



The trend appears to be flat overall, varying between about \$40,000 and \$60,000 with consistent peaks near January and consistent lows near August.

This is a simple example of how you can use a chart to visualize data in a spreadsheet. You'll learn more about creating, modifying, and interpreting data visualizations in an upcoming lesson.

Step 5: Adjust the chart

Use the **Chart editor** panel to make changes to the chart. In the **Setup** tab, you can change the chart type, data range, axis labels, and more. For example, the chart type drop-down list changes the type of chart from line chart to a different style, such as a bar chart. In the **Customize** tab, you can change many other options, such as the chart's style.

If the **Chart editor** panel isn't open, double-click on the chart to open it. Then, change the various components of the chart by selecting them within the chart or by navigating to them from the **Chart editor** panel.

In this example, change the chart title to "Monthly sales." Follow these steps:

1. If necessary, open the **Chart editor** panel by double-clicking on the chart.
2. In the **Chart editor** panel, select the **Customize** tab.
3. Expand the **Chart & axis titles** section.
4. Verify that the **Chart title** is selected from the dropdown, then change the title to **Monthly sales**. Your chart will now display the title "Monthly sales"
5. If you're using Microsoft Excel, double-click on the title in the chart to edit directly.

Get creative! Don't be afraid to play around with the options. You can always save or download another copy of the template if you want to start fresh!

Pro Tip: Save the activity spreadsheet template

Be sure to save a copy of the spreadsheet template you used to complete this activity. You can use it for further practice or to help you work through your thought processes for similar tasks in a future data analyst role.

1.

Question 1

Reflection

Based on the chart you created, when is the busiest time of year for your company?

- ☐ September-November
- ☐ March-May
- ☒ December-February
- ☐ June-August

✓ **Correct**

The busiest time of year is December-February.

2.

Question 2

Examine your completed data visualization. In the space provided below, write 2-3 sentences (40-60 words) in response to each of the following questions:

- How does the line chart help you visualize the sales data?
- Based on your analysis, when will your company need to increase staff and inventory?

Answer

- When I looked at the graph, I noticed that in December to February, the total sales peak is high, and the June to September graph is the lowest. that define the complete picture of company total sales, which gives me the thorough insights. which means in the December to February company total sales are highest in the year and June to September total sales are lowest in the year.
- In December to February, total sales are highest for the company. This time the company needs to increase staff and inventory.

Correct

Congratulations on completing this hands-on activity! An effective response would include a discussion of how visualizing the data lets you determine that your company has its highest sales during the months of December through February. During these months, your company will need to increase its on-hand inventory and hire temporary workers to help with the increased sales during tourist season.